

Byung-Hak Hwang

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Research interests

- Algebraic combinatorics and related topics.
- Formalization of mathematics.
- AI for mathematics.

Education and Employment

- **Research Fellow, Korea Institute for Advanced Study**
May 2024 –
Mentor: Hyun Kyu Kim
- **Alternative Military Service**
April 2021 – April 2024
- **Research Fellow, Applied Algebra and Optimization Research Center**
September 2020 – March 2021
Mentor: Jang Soo Kim
- **Ph.D. Mathematics, Seoul National University**
March 2014 – August 2020
Advisor: Woong Kook
- **B.S. Double major in Mathematics, and Electric & Computer Engineering, Seoul National University**
March 2009 – February 2014

Publications

Journal publications and preprints

10. *Composition Direction of Seymour's Theorem for Regular Matroids – Formally Verified* (with M. Dvorak, T. Figueroa-Reid, R. Hamadani, E. Karunus, V. Kolmogorov, A. Meiburg, A. Nelson, P. Nelson, M. Sandey, I. Sergeev.)
preprint, arXiv:2509.20539, submitted.
9. *Refinement of Hikita's e -positivity theorem via Abreu–Nigro's g -functions and restricted modular law* (with J. Huh, D. Kim, J. S. Kim, J. Oh)
preprint, arXiv:2504.09123, submitted.
8. *Refined canonical stable Grothendieck polynomials and their duals, Part 2* (with J. Jang, J. S. Kim, M. Song, U-K. Song)
European Journal of Combinatorics, 127:Paper No. 104166, 34, 2025.

7. *Noncommutative symmetric functions and skewing operators*
Discrete Mathematics, 348(1):Paper No. 114255, 2025.
6. *Refined canonical stable Grothendieck polynomials and their duals, Part 1* (with J. Jang, J. S. Kim, M. Song, U-K. Song)
Advances in Mathematics, 446:Paper No. 109670, 42, 2024.
5. *Chromatic quasisymmetric functions and noncommutative P -symmetric functions*
Transactions of the American Mathematical Society, 377(4):2855–2896, 2024.
4. *A combinatorial model for the transition matrix between the Specht and SL_2 -web bases* (with J. Jang and J. Oh)
Forum of Mathematics, Sigma, 11 (2023), E82.
3. *Acyclic orientation polynomials and the sink theorem for chromatic symmetric functions* (with W.-S. Jung, K.-J. Lee, J. Oh, and S.-H. Yu)
Journal of Combinatorial Theory, Series B, 149 (2021), 52-75.
2. *On linearization coefficients of q -Laguerre polynomials* (with J. S. Kim, J. Oh, and S.-H. Yu)
The Electronic Journal of Combinatorics, Volume 27, Issue 2 (2020), P2.22.
1. *Reverse plane partitions of skew staircase shapes and q -Euler numbers* (with J. S. Kim, M. Yoo, and S.-m. Yun)
Journal of Combinatorial Theory, Series A, 168 (2019), 120-163.

Conference proceedings

6. *Refined canonical stable Grothendieck polynomials and their duals* (with J. S. Kim, J. Jang, M. Song, and U-K. Song)
Proceedings of FPSAC 2023.
5. *Chromatic quasisymmetric functions and noncommutative P -symmetric functions*
Proceedings of FPSAC 2023.
4. *A combinatorial model for the transition matrix between the Specht and web bases* (with J. Jang, and J. Oh)
Proceedings of FPSAC 2022.
3. *Acyclic orientation polynomials and the sink theorem for chromatic symmetric functions* (with W.-S. Jung, K.-J. Lee, J. Oh, and S.-H. Yu)
Proceedings of FPSAC 2020.
2. *On linearization coefficients of q -Laguerre polynomials* (with J. S. Kim, J. Oh, and S.-H. Yu)
Proceedings of FPSAC 2020.
1. *Reverse plane partitions of skew staircase shapes and q -Euler numbers* (with J. S. Kim, M. Yoo, and S.-m. Yun)
Proceedings of FPSAC 2018.

Presentations

- Formalizing mathematics: why and how, HCMC Quantum Analysis & Geometry, August 2025
- Refinement of Hikita’s e -positivity theorem via Abreu–Nigro’s g -functions and restricted modular law, AlCoVE 2025, online, May 2025

- Formalizing mathematics: why and how, Korea University, May 2025
- Formalizing mathematics: why and how, 2025 KMS Annual Meeting, KAIST, April 2025
- Formalizing mathematics: why and how, Sungkyunkwan University, March 2025
- Noncommutative symmetric functions and skewing operators, The 33rd KIAS Combinatorics Workshop, KIAS, February 2025
- Braid varieties, R -polynomials, and rational Catalan number, Combinatorics on flag varieties and related topics 2025, Ajou University, February 2025
- Formalizing mathematics: why and how, Combinatorics Seminar for Young Researchers, Inha University, January 2025
- Cluster combinatorics of Grassmannians, Intensive workshop on algebraic combinatorics, January 2025
- Formalizing mathematics: why and how, AI for Mathematics Workshop on Formalization, KIAS, December 2024
- Formalizing mathematics: why and how, Hanyang University, December 2024
- Braids and R -polynomials, 2024 KMS Annual Meeting, Sungkyunkwan University, October 2024
- Formalizing mathematics: why and how, Workshop on Mathematical Challenges 2024, KIAS, October 2024
- The theory of noncommutative symmetric functions, Topology and Combinatorics seminar at Ajou University, June 2024
- Logarithmic concavity of Kazhdan–Lusztig $\tilde{R}$ -polynomials, Workshop for Young Representation Theorist in Korea, December 2023
- Chromatic quasisymmetric functions and noncommutative P -symmetric functions, FPSAC 2023, July 2023
- Chromatic quasisymmetric functions and noncommutative P -symmetric functions, Combinatorics on flag varieties and related topics 2023, February 2023
- A proof of Harada–Precup’s conjecture, SKKU Combinatorics seminar, November 2020
- Acyclic orientation polynomials, 2020 KMS Annual Meeting, October 2020
- Positivity in symmetric functions, AORC Monthly seminar, September 2020
- Chromatic quasisymmetric functions and noncommutative P -symmetric functions, Topology and Combinatorics seminar at Ajou University, September 2020
- Acyclic orientation polynomials, 2020 Combinatorics Workshop, August 2020
- Acyclic orientation polynomials and the sink theorem for chromatic symmetric functions (poster), The 32nd international conference on Formal Power Series and Algebraic Combinatorics (online), July 2020
- Reverse plane partitions of skew staircase shapes and a modification of Lindström–Gessel–Viennot lemma, The 18th KIAS Combinatorics Workshop, 2017